

CSC3210 - Quiz - System Calls and Processes

Name: Solution

Answer the questions in the spaces provided. Work is to be done on your own without a computer/phone. Justify your answers and show all work where appropriate.

1. (10 points) What is the difference between a program and a process? Provide an example of how multiple processes can be created from the same program.

Solution: A program is a static file containing code, while a process is a program in execution, which has a life cycle and changes its data during execution.

For example:

- the executable for the 'Word' word processor is an example of a program.
- multiple process instances of the 'Word' program can all be running from the same program each working on their own different document.

2. (10 points) What is the difference between user mode (space) and kernel/system mode (space)? Why would it not be a good idea to allow a user program to execute in kernel/system mode?

Solution: The difference is in CPU instruction privilege. The CPU is not able to execute certain instructions in user mode which restricts applications from accessing certain hardware features and memory address ranges. In kernel/system mode, the CPU can execute all instructions and has full access to all hardware and memory. A user program with access to all hardware and memory can access other processes running on the system and the OS itself which could cause a security risk or a safety risk.

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3. (10 points) What hardware-supported mechanism allows a process to safely transition from user mode to kernel mode when making a system call? Describe how it ensures protection.

Solution: The mechanism used to safely switch from user mode to system mode is called a trap, which is a software interrupt that invokes the operating system code to handle the request and change the mode. It ensures protection because user code cannot switch to kernel mode on its own; only specific CPU instructions can change privilege levels. The entry addresses for system calls are stored in memory only the kernel can modify.

4. (10 points) In the 5 process state model one of the states is “waiting”. Describe the “waiting” state and provide one advantage of having waiting in addition to the other states. Why would a process be in a state of “waiting”?

Solution: The waiting state is the state a process enters when it cannot continue executing until some event occurs. A process in the waiting state is not eligible to run because it is waiting for something external. Using this state prevents the operating system from attempting to run a process that will not do any useful work when it is waiting for something (I/O, another process, etc.).